ECON3700
ENVIRONMENTAL ECONOMICS

Analytical Tools II

Economics of Environmental Quality &

Criteria for Evaluating Policy

References: Chapter 5 & 9, ISE Environmental Economics, 9th Edition

Presenter: Dr Ammarr Aripin, School of Economics.

Learning Objectives

1. Explain environmental damages, their types, and how they are represented through emission and ambient damage functions.
2. Interpret marginal and total damages, including the effects of uncertainty and irreversibility.
3. Define abatement, describe common methods, and analyse abatement cost functions.
4. Aggregate abatement costs across multiple sources to achieve least-cost pollution control.
5. Determine the socially efficient level of emissions using MAC–MD analysis and calculate total damages and abatement costs.
6. Evaluate environmental policies using criteria such as efficiency, cost-effectiveness, fairness, enforceability, innovation incentives, and other relevant principles.



Part A: Damages

Pollution Control

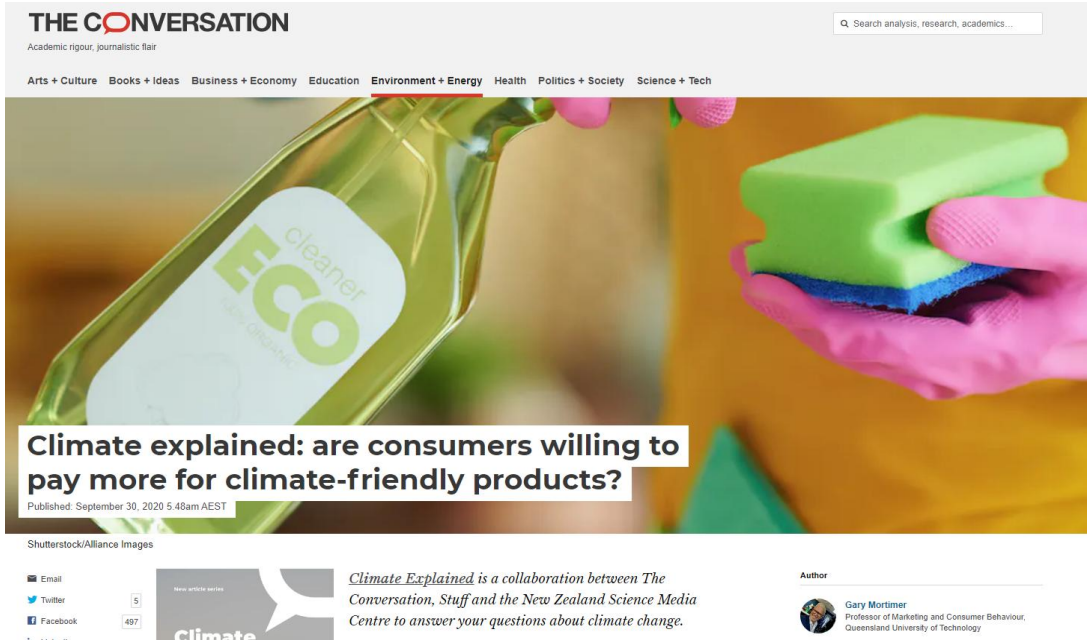
If markets lead to an **inefficient** level of pollution,

what is an efficient level of pollution?

Efficiency means the right trade-off between:

WTP: willingness to pay to avoid damages of pollution

MC: marginal cost of reducing emissions of pollution

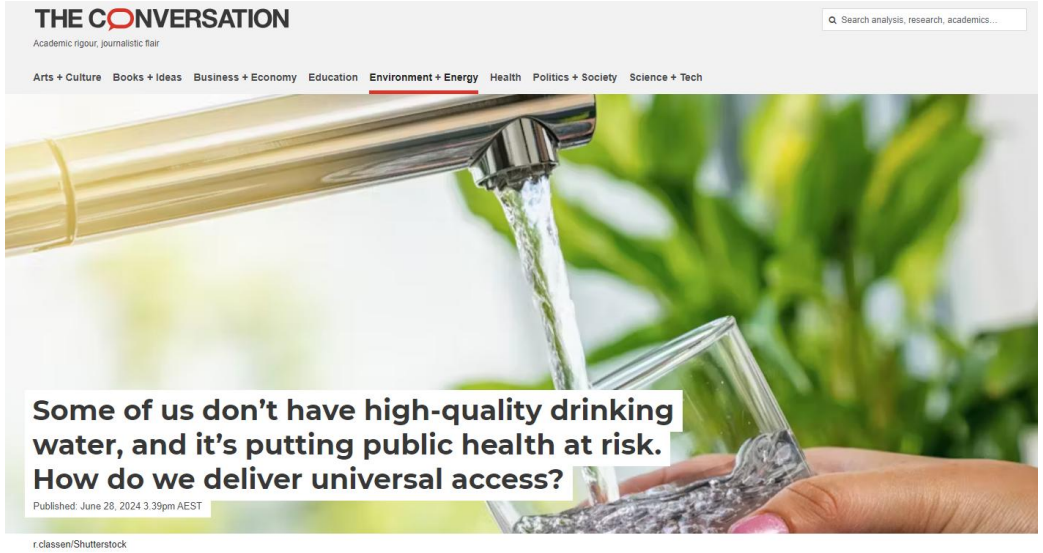


Source: [Climate explained: are consumers willing to pay more for climate-friendly products? (theconversation.com)]

Damages

Damages are **negative** impacts of environmental degradation:

- Human health impacts e.g., *waterborne disease, asthma*
- Recreation impacts e.g., *reduced enjoyment of polluted national parks*
- Impacts on non-use values e.g., *extinction of species*

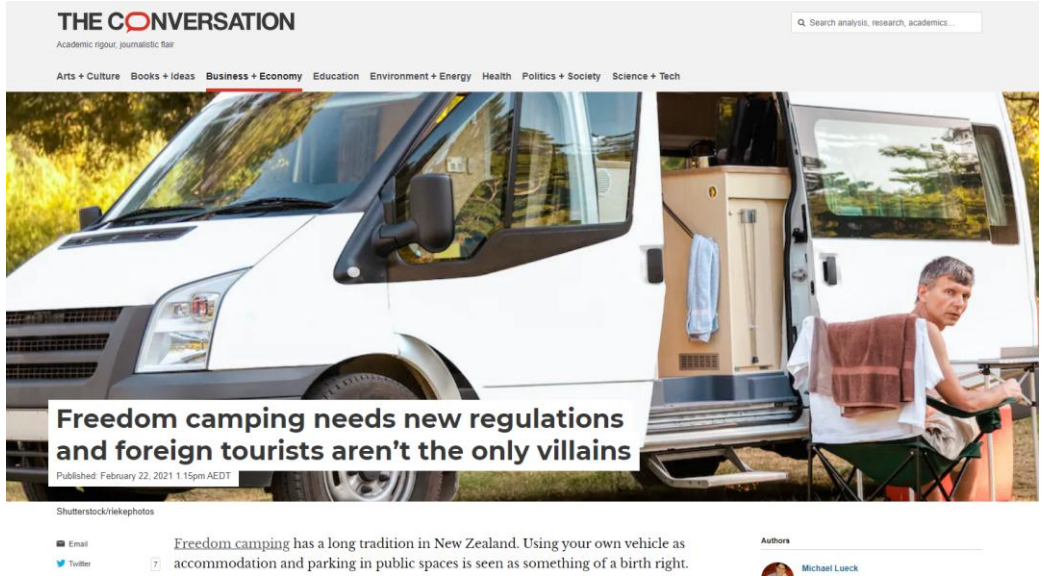


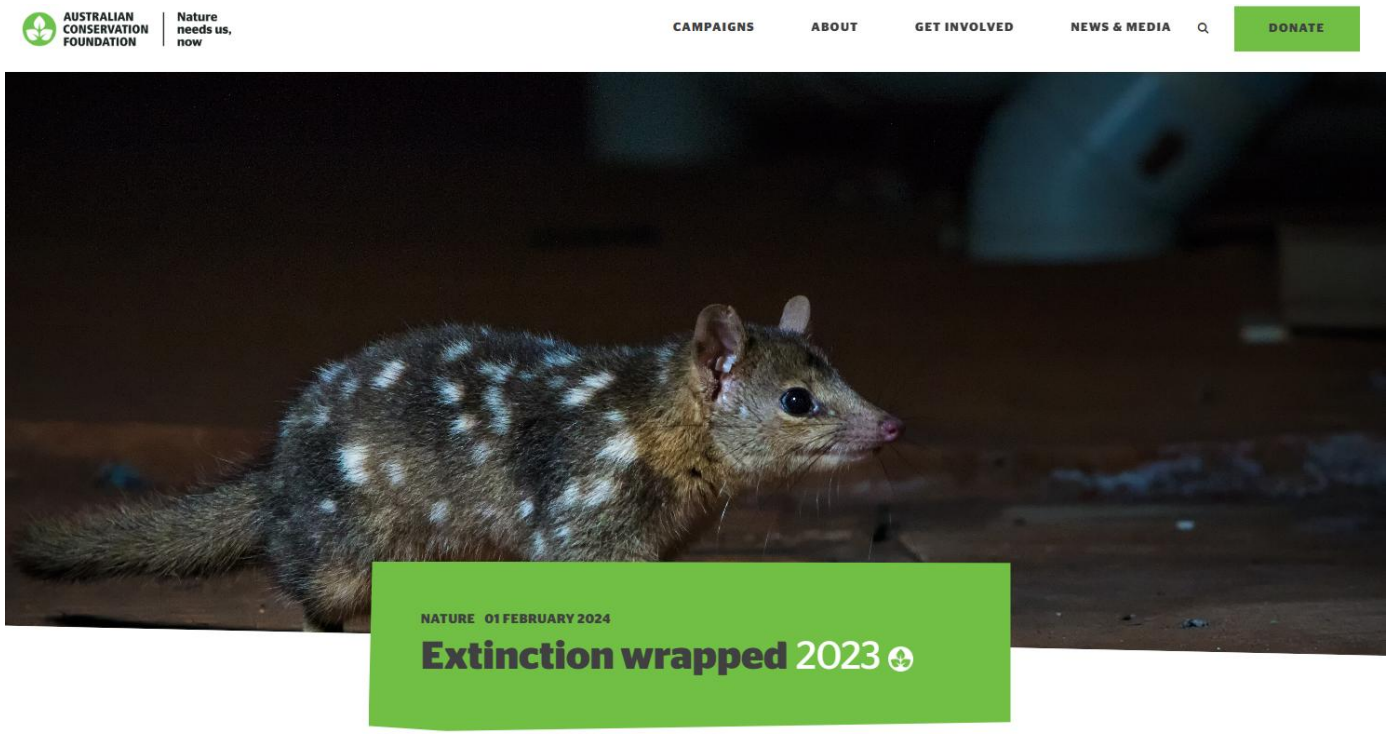
Source:

[Some of us don't have high-quality drinking water, and it's putting public health at risk. How do we deliver universal access? (theconversation.com)]

Source:

[Freedom camping needs new regulations and foreign tourists aren't the only villains (theconversation.com)]

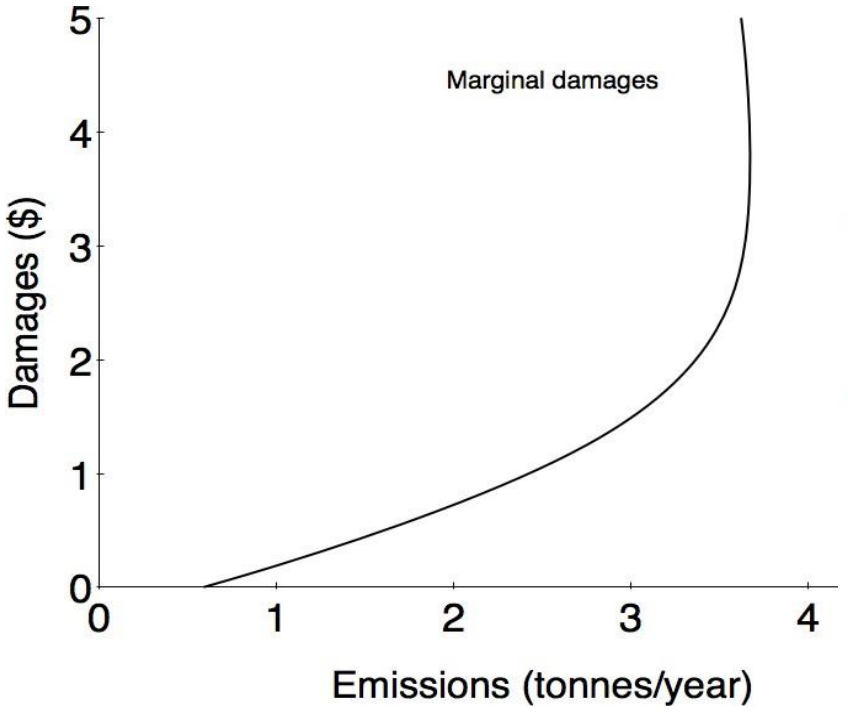


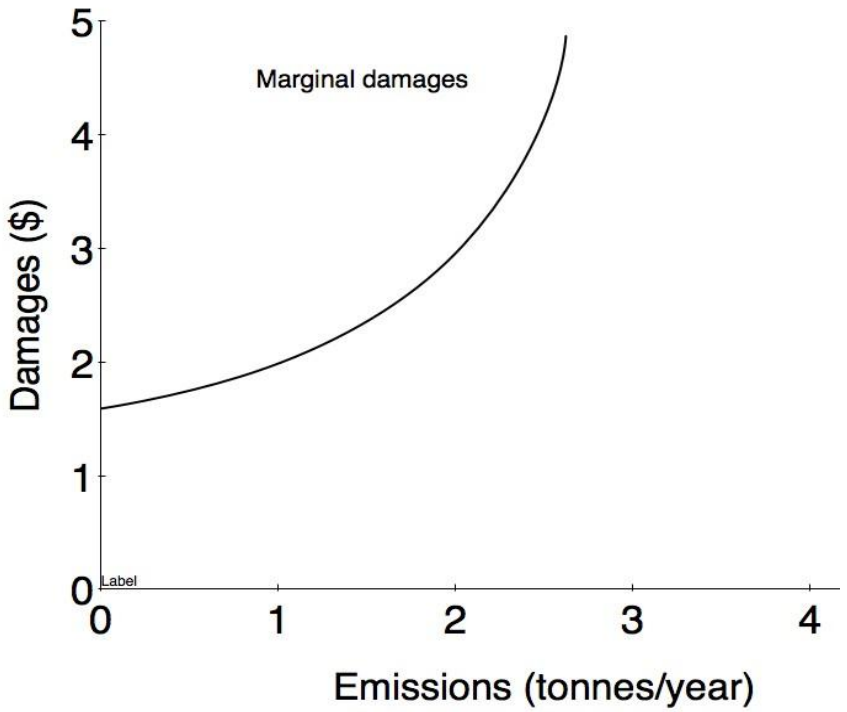


Source: [Extinction wrapped 2023 - Australian Conservation Foundation (acf.org.au)]

Damage Functions

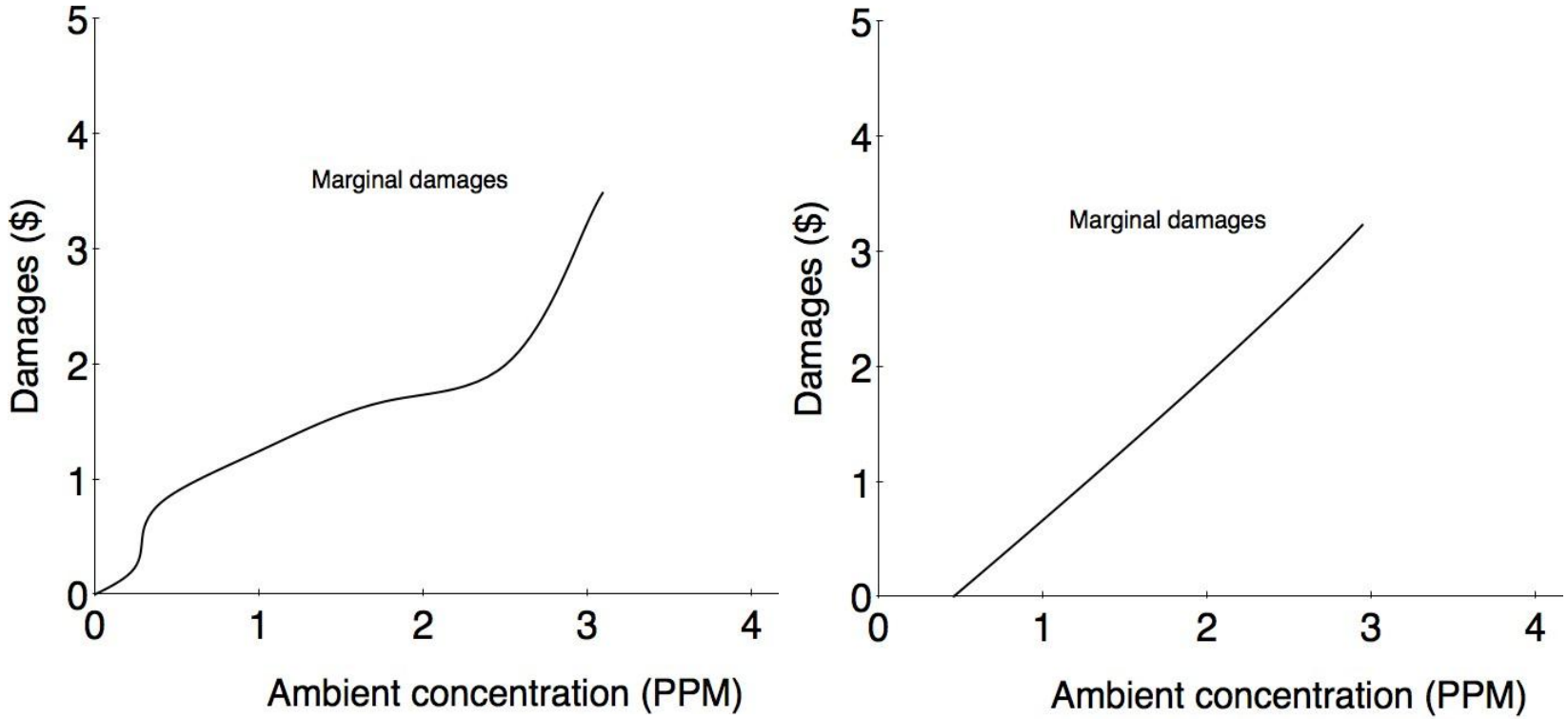
Damage functions describe the relationship between emissions and impacts.





Damage Functions

Damage functions describe the relationship between ambient concentration levels and impacts.



Emission VS Ambient Damage Functions

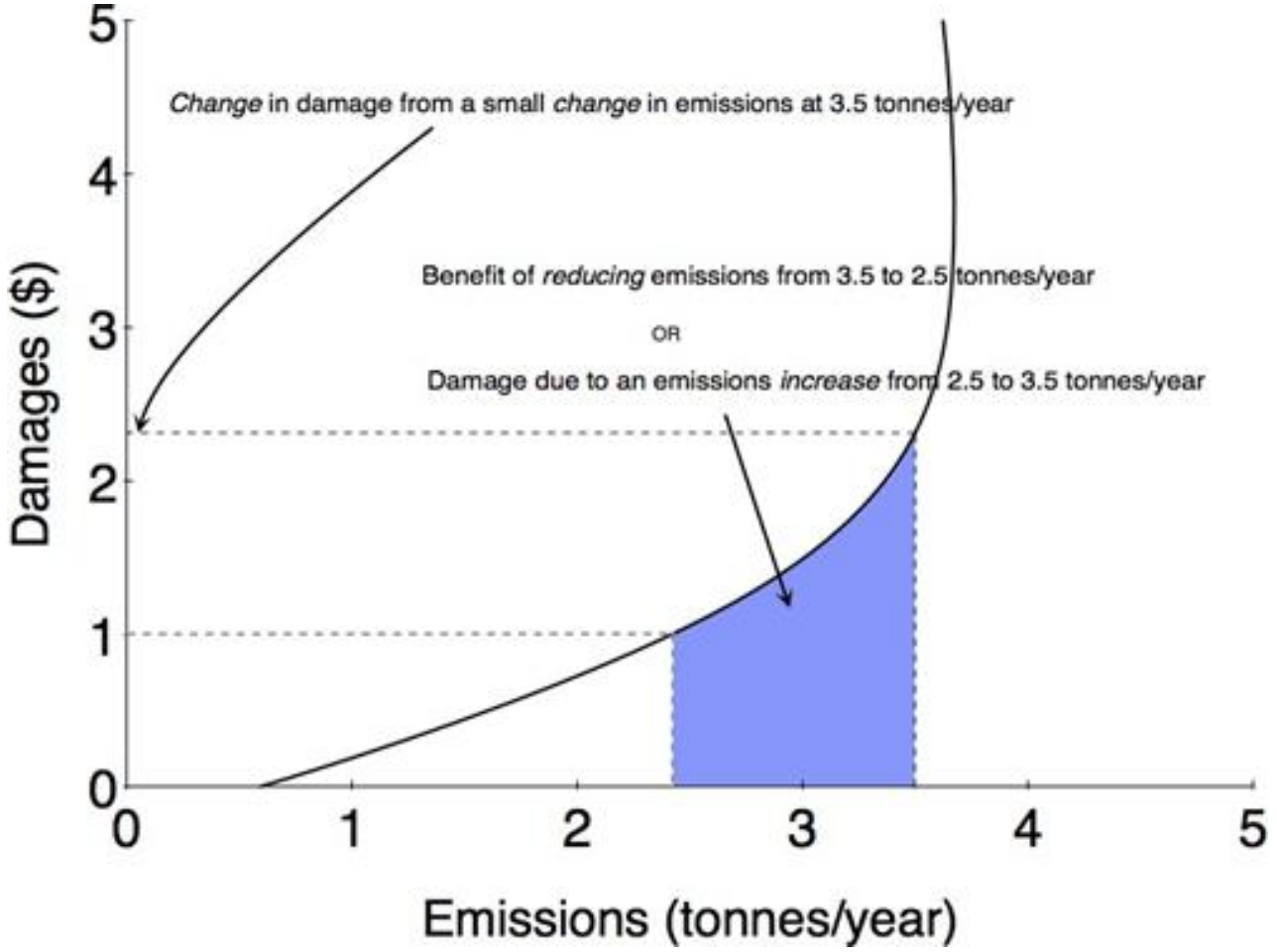
Emission damage functions:

relationship between quantity emitted and damages.

Ambient damage functions:

relationship between concentration of a pollutant in the ambient environment and the resulting damages.

Marginal VS Total Damage



Damages, Technology & Uncertainty

For the same level of emissions/concentrations, damages may differ because of proximity to populations or local preferences.

Damage functions are not known with certainty: they may change in response to new scientific information (more or less damage from same emissions level).

Damages may also be irreversible (cannot reduce damages by reducing emissions).



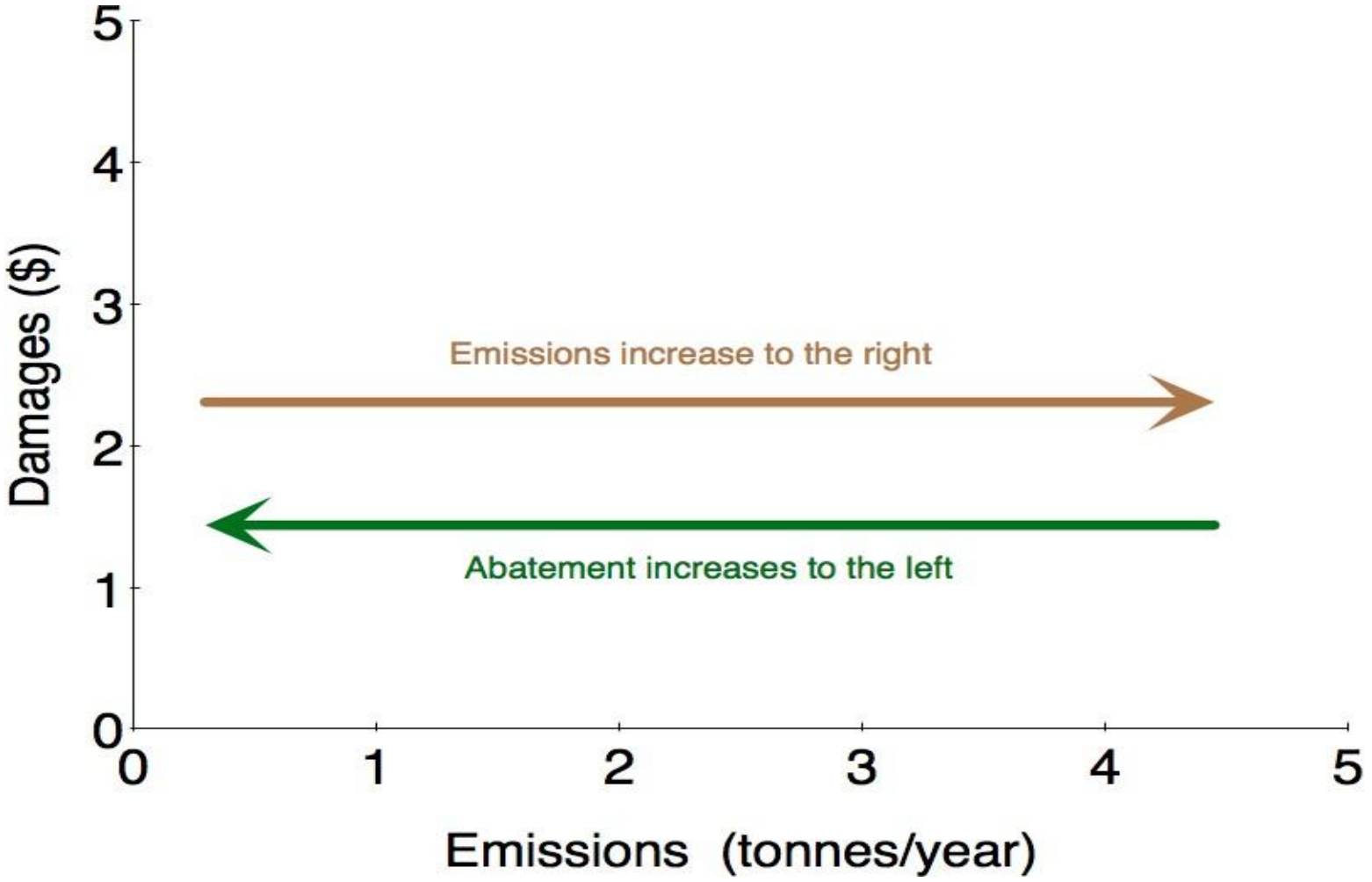
Part B: Abatement

Abatement

Abatement means reducing emissions or lowering ambient concentrations.

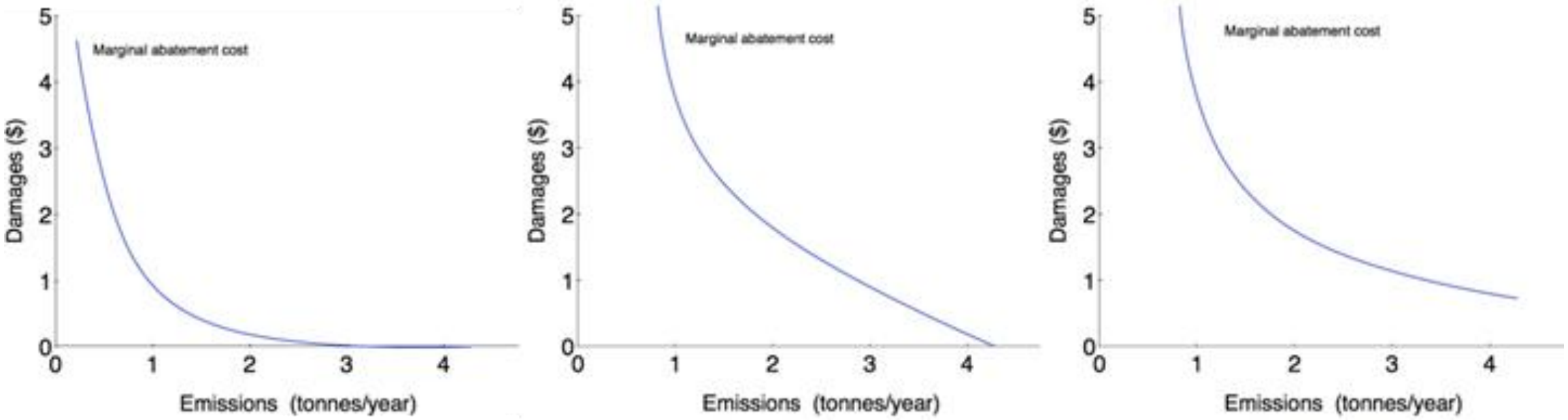
Example; using different inputs, installing more efficient technology, recycling residuals, treatment of residuals or from plant shutdown.

Abatement


Damages ($)
5
4
3
2
1
0
Emissions increase to the right
Abatement increases to the left
0
1
2
3
4
5
Emissions (tonnes/year)

Abatement Cost Functions

Unfortunately, abatement is not usually free.

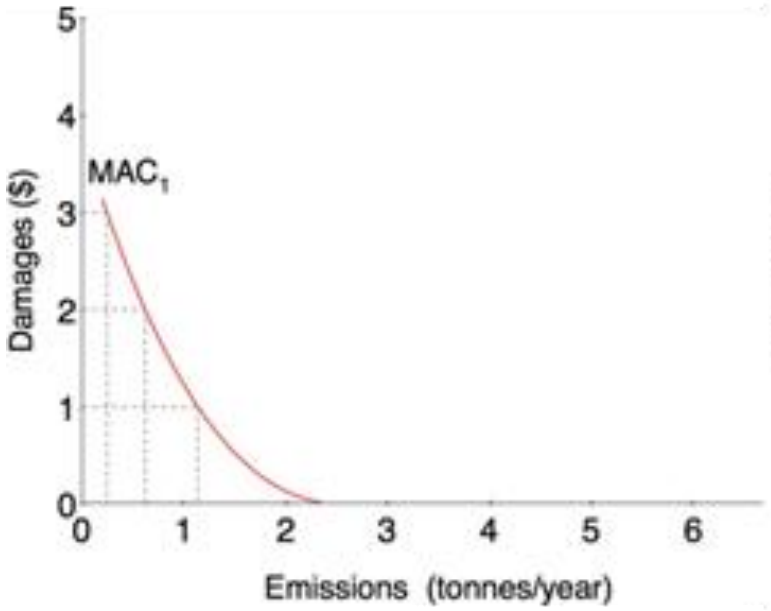


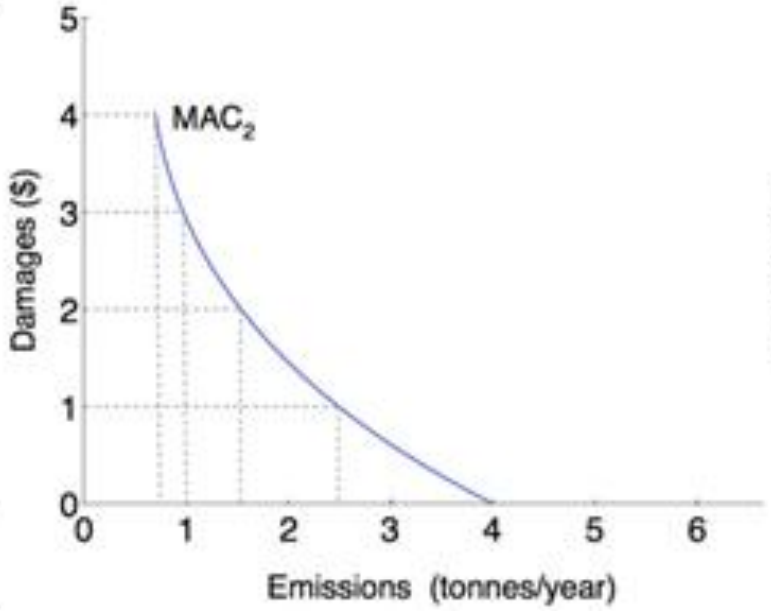
Aggregating Abatement Costs

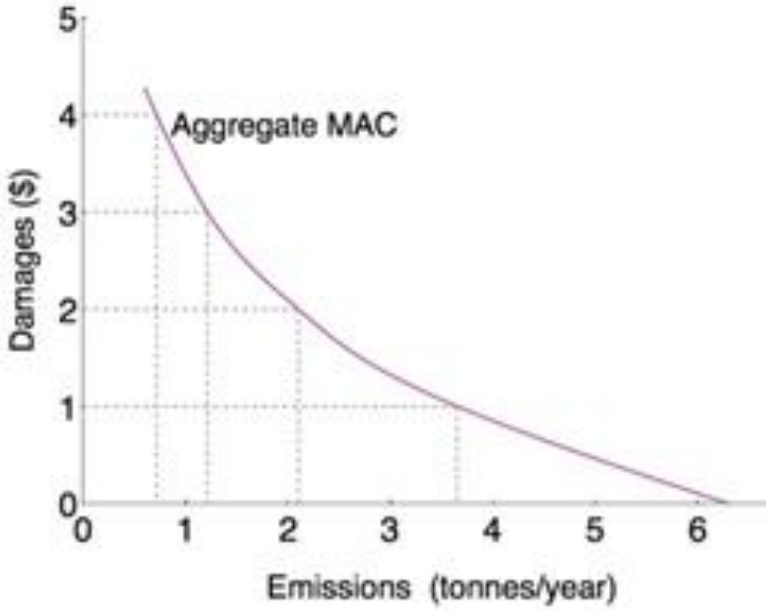
- Usually more than one firm/plant produces emissions and can undertake abatement.
- Often different sources have different abatement costs.
- To work out what level of emissions is optimal, we want to aggregate emissions across firms/plants.
- We want to aggregate in a least cost manner: we implicitly assume that the lowest cost sources of abatement are the first sources that are “switched on”.

Abatement Cost Functions

To aggregate marginal abatement cost curve, add horizontally for each level of cost.



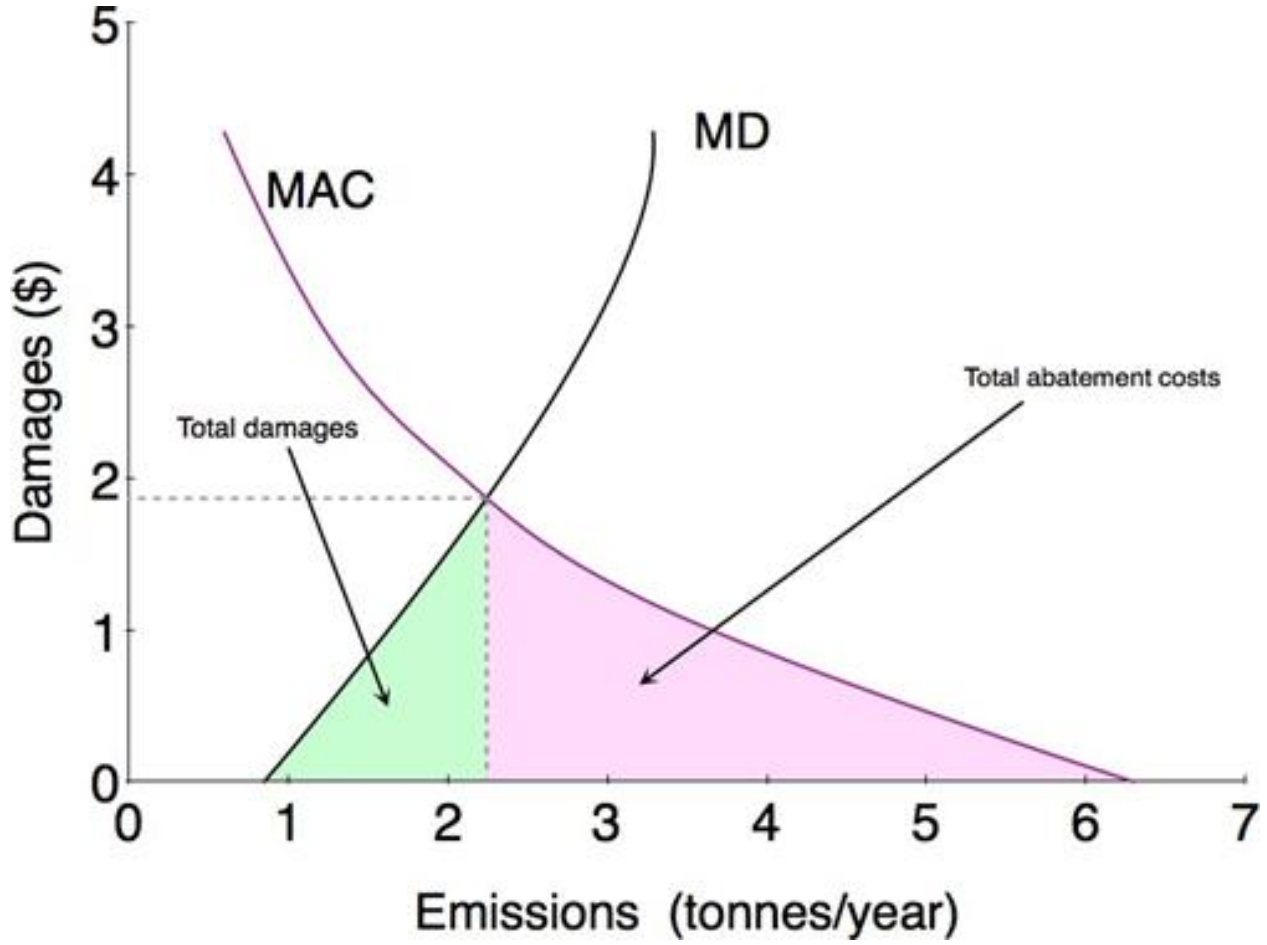




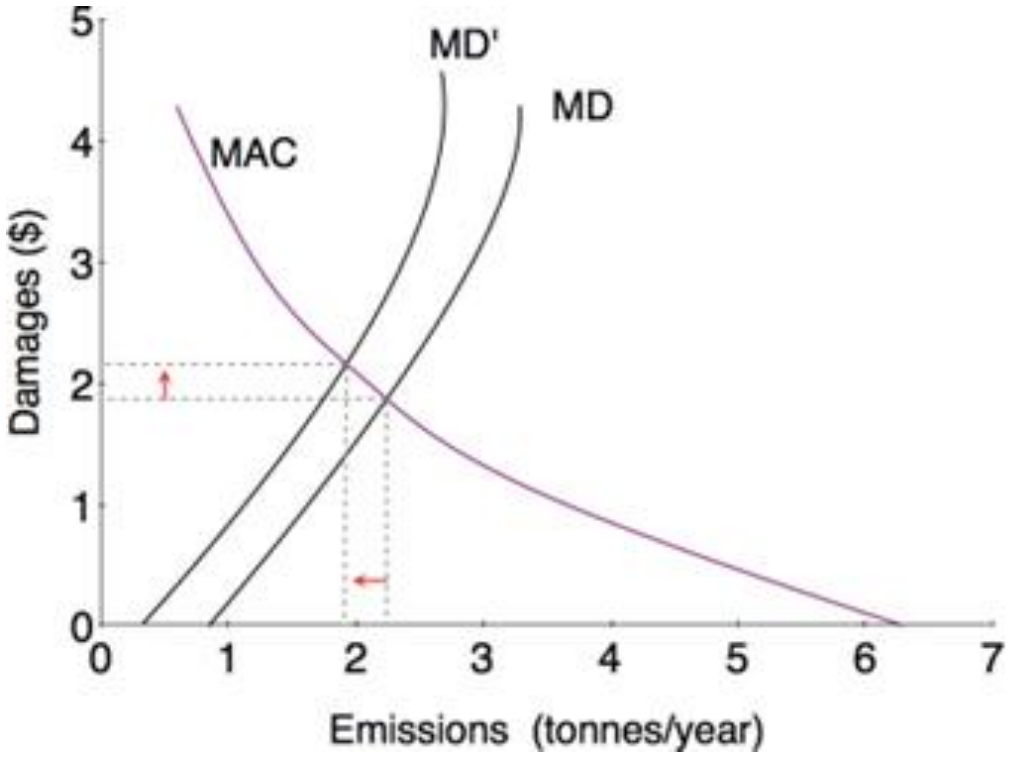


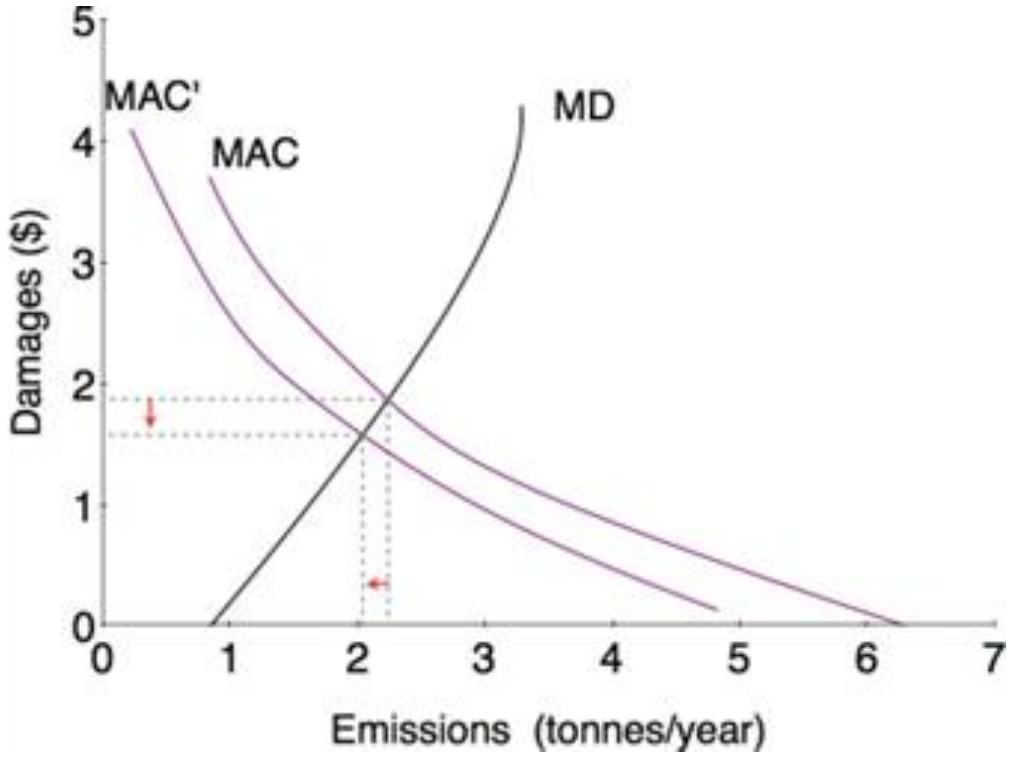
Part C: Social Optimality

The Efficient Level of Emissions



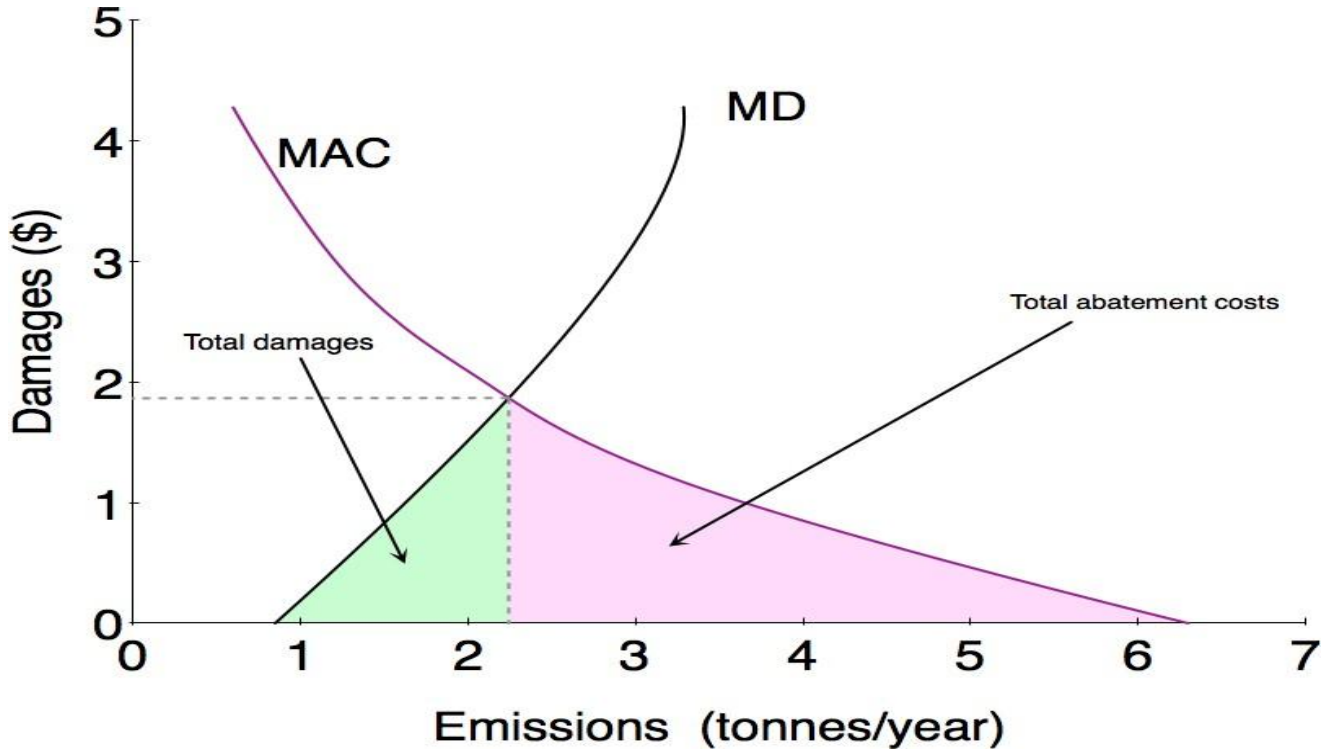
Changes in the Efficient Level of Emissions



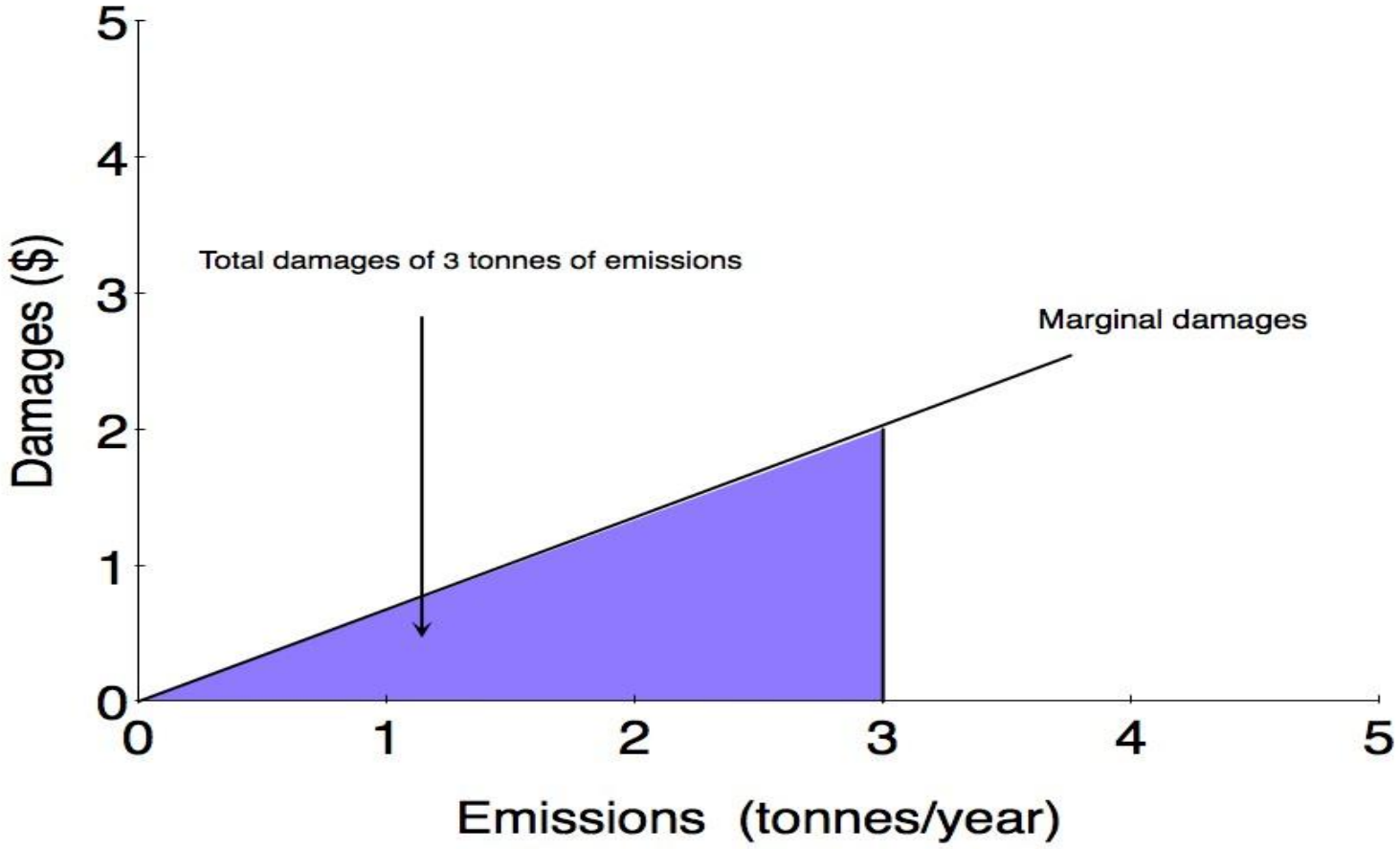


Calculating Totals

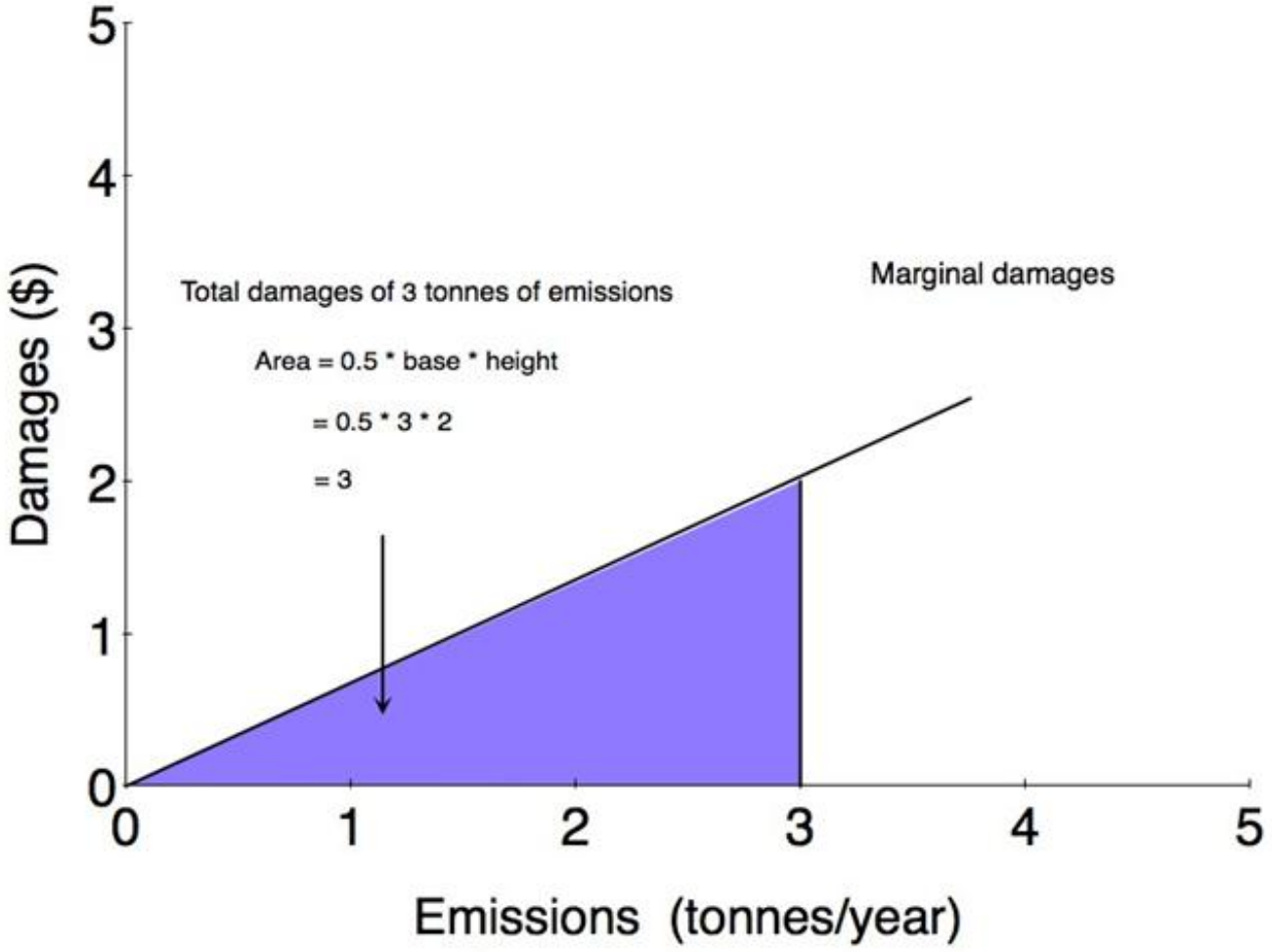
Recall that total damages (total abatement costs) are the area under the marginal curve.



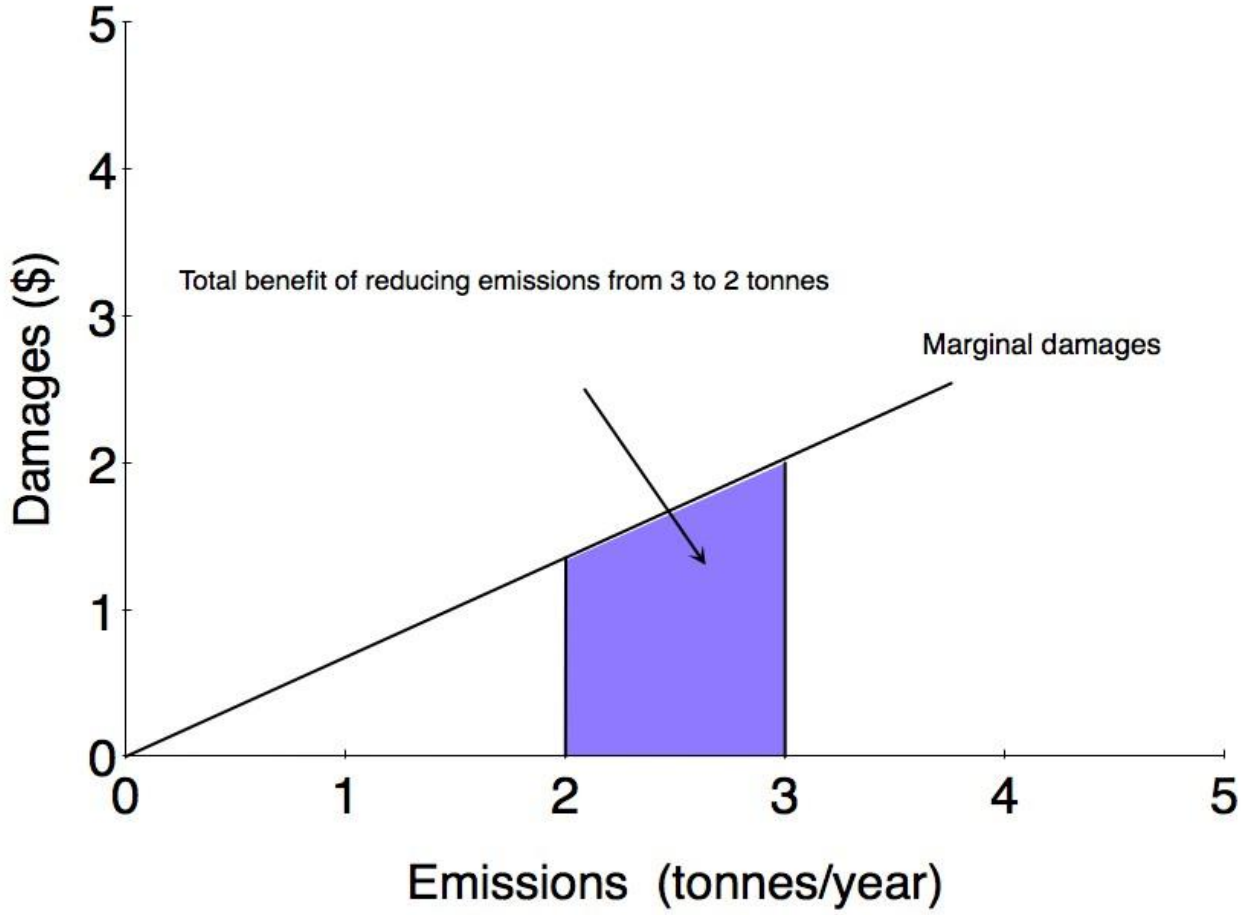
Calculating Total Damages



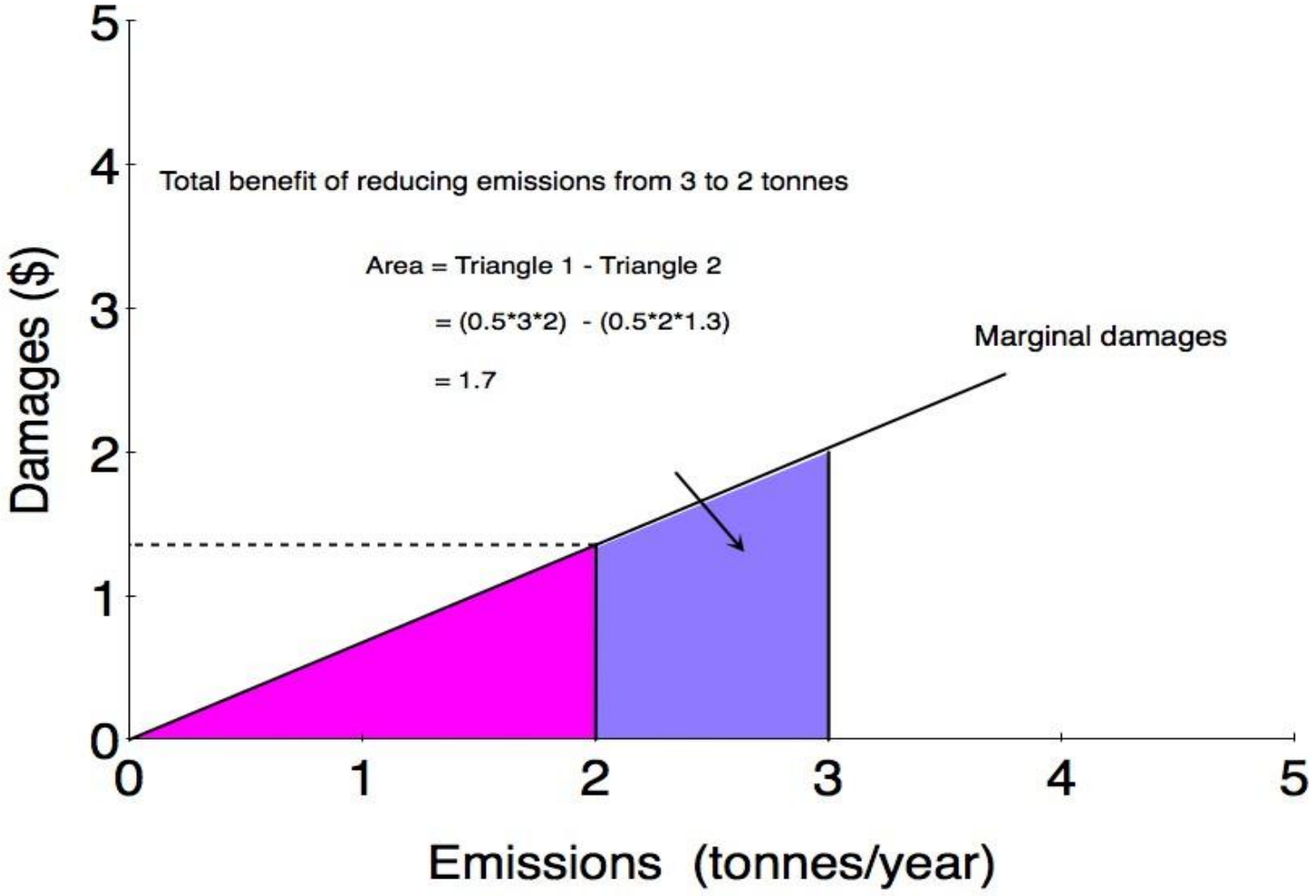
Calculating Total Damages



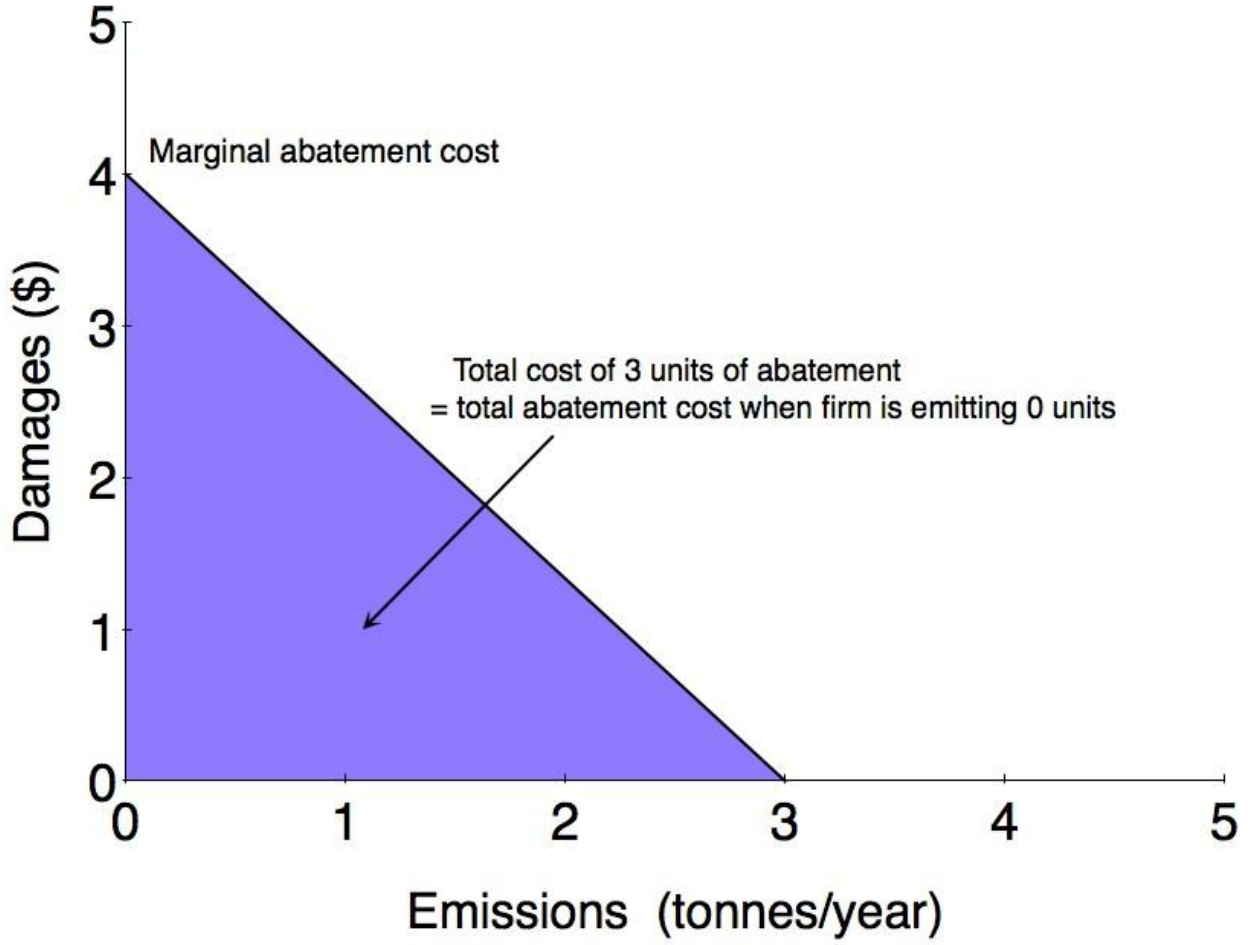
Calculating Benefits of Emissions Reductions



Calculating Benefits of Emissions Reductions

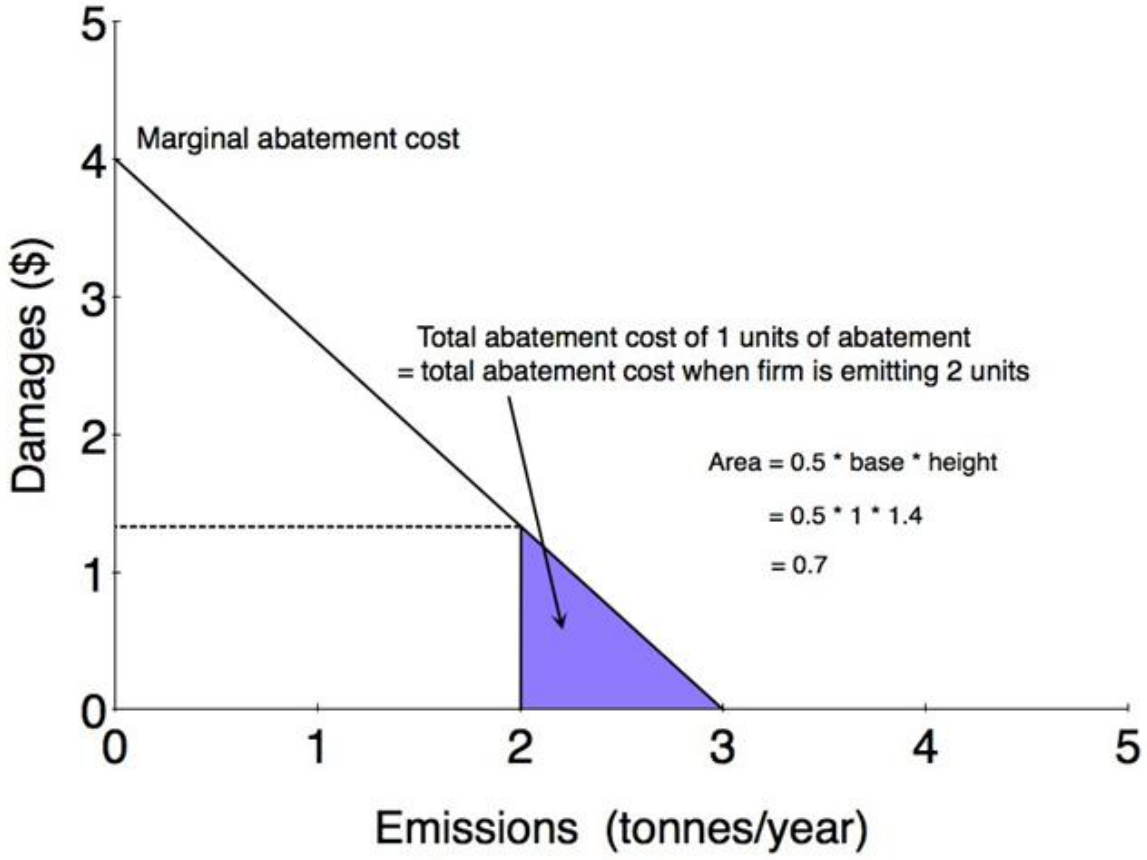


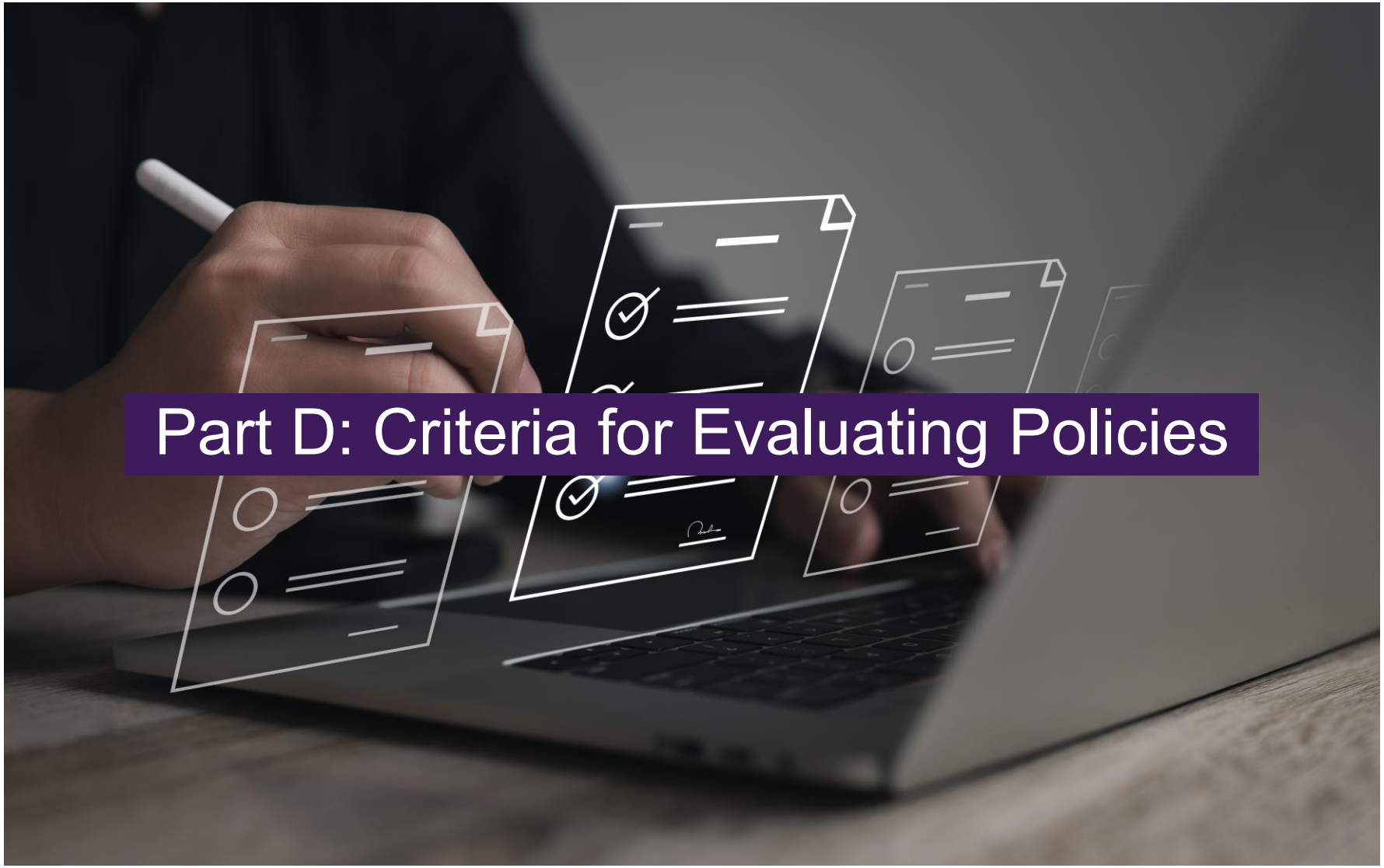
Calculating Total Abatement Costs



Calculating Total Abatement Costs

How do I calculate the total abatement cost of 1 unit of abatement?





Part D: Criteria for Evaluating Policies

Criteria for Evaluating Policies

- Efficiency and Cost Effectiveness
- Fairness / Equity
- Enforceability
- Incentives for Technological Innovation (Dynamic Efficiency)
- Other

Efficiency

Efficiency: minimizes the total social costs of pollution
How close is the outcome to where MAC=MD?
Are TAC minimised?

Cost-Effectiveness

Either:

- Maximise environmental outcomes from given budget
- Minimise cost to achieve an environmental target

An efficient policy is cost-effective but not necessarily vice versa

Fairness

Does policy improve or worsen the distribution of damages from pollution?

Are the costs of the policy disproportionately borne by one segment of the population?

Enforceability

Enforcement is crucial for ensuring compliance with regulations but is costly.

- Monitoring: measuring the performance of polluters relative to legal requirements.
- Sanctioning: penalising those who are found in violation.
- Probability of compliance = f(likelihood of detection, sanction if detected).

Incentives for Technological Innovations

Does the policy provide a strong incentive to find new innovative ways of reducing emissions?

Dynamic efficiency = shifting the MAC down over time

Other Criteria?

- Flexibility
- Risk
- Certainty
- Polluter pays
- Precautionary principle